How-To Library

Informatica

Ingest Metadata from Custom Resources

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Abstract

Enterprise Data Catalog, by default, provides system models for multiple data sources from which you can extract metadata. If you want to ingest custom metadata in the catalog, you need to define a custom model and create a custom resource. This article explains how to ingest custom metadata from a data source for which Enterprise Data Catalog does not provide a resource.

Supported Versions

- Informatica Enterprise Data Catalog 10.4.1

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Overview

You can use custom metadata integration to ingest metadata from data sources for which Enterprise Data Catalog does not provide a resource. Based on the model for the data source, Enterprise Data Catalog extracts metadata from the data source.

Custom metadata integration involves the following tasks:

1. Create the custom model
2. Create the custom resource type
3. Create the custom resource

Tutorial Example

You are an application owner in the retail industry, and data analysts in your organization need to prepare a report on the quarterly sales. The information that is required to prepare the report is stored in Microsoft Access Database. You are asked to import metadata from the Microsoft Access Database into Enterprise Data Catalog.

The following table lists the challenge you might come across and the possible solution that you achieve using Enterprise Data Catalog:

Challenge	Solution
Enterprise Data Catalog does not provide a resource for Microsoft Access Database.	You can use custom metadata integration to ingest metadata from the Microsoft Access Database into the catalog.

In this tutorial, you will complete the following lessons to ingest metadata from Microsoft Access Database into the catalog:

Lesson	Description
Create a custom model	Create a custom model for Microsoft Access Database in Catalog Administrator.
Create a custom resource type	Create the **Access Database** resource type and associate it with the **com.example.acessdb** custom model.
Export the custom resource type template	Export the **Access Database** export template to add the object details of a specific class defined in the model.
Enter the association details	Enter the association details in the object.csv and class details in links.csv file to prepare the template .zip file.
Create the custom resource	Create the **Reporting_AccessDB** custom resource using the **Access Database** custom resource type.
View the custom metadata resource and assets	Search the **Reporting_AccessDB** resource and view asset details in Enterprise Data Catalog application.

Lesson 1: Create the Custom Model

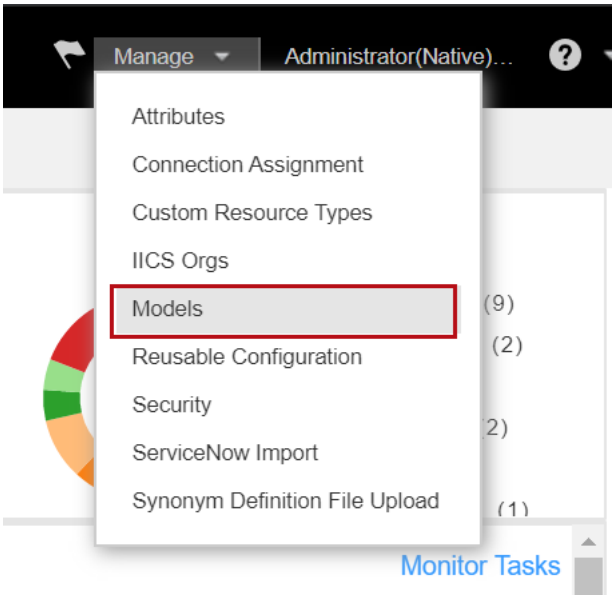
You can define a custom model in Catalog Administrator. The model defines the structure of the metadata stored in the catalog.

As a one-time step, use model definition files that Enterprise Data Catalog provides in the XSD format to create and validate the .xml files. For more information about model definition files, see the Downloading the Model Definition Files for Validation topic.

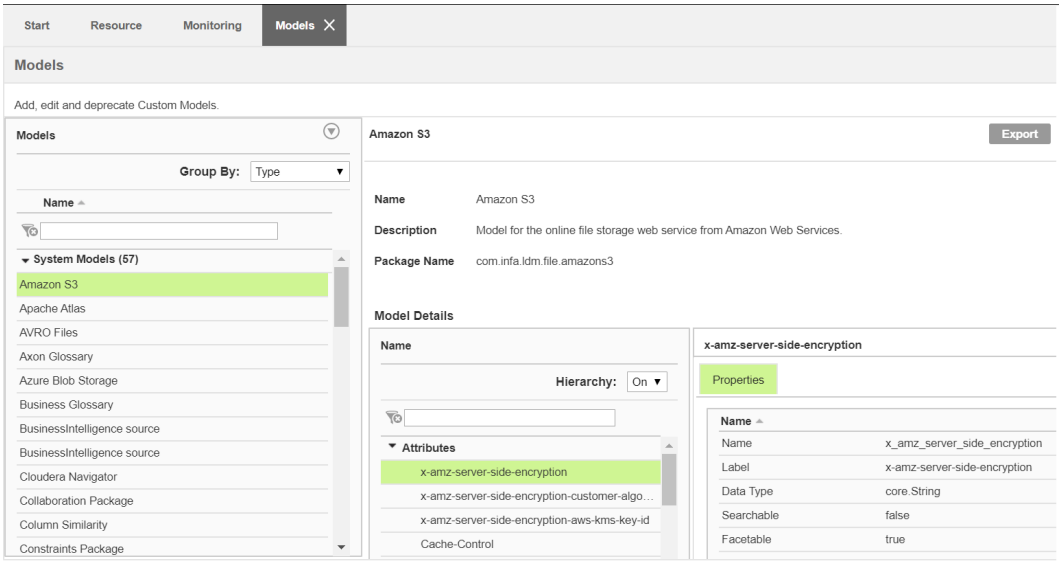
In this lesson, we will create a custom model for Microsoft Access Database. Before you create a custom model, make sure that you have created and validated the model .xml file.

1. Log in to the Catalog Administrator.

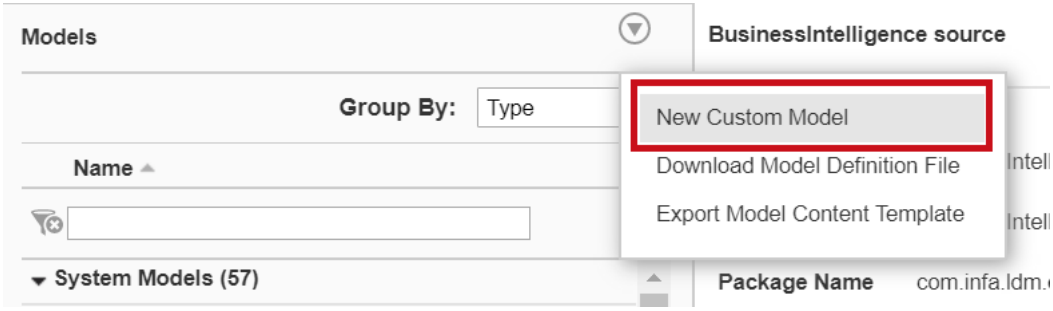
2. In the Catalog Administrator header area, click **Manage** > **Models**.



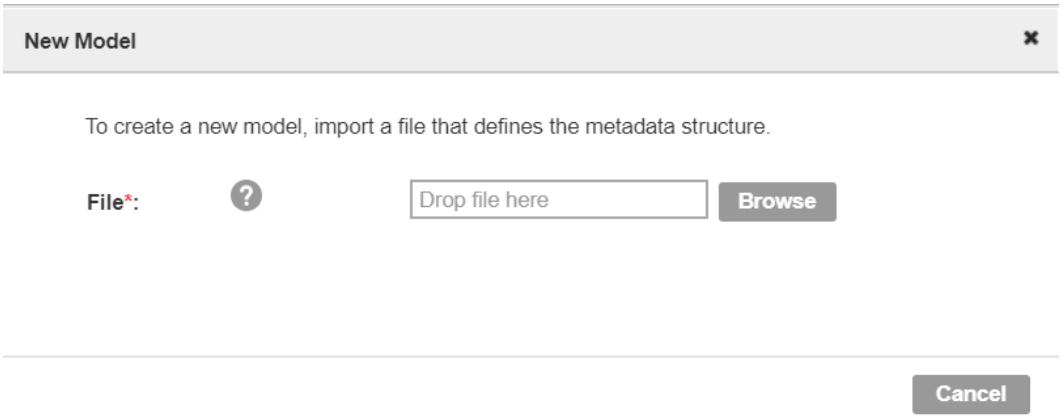
The **Models** page appears.



3. Click the ( ) icon from the **Models** pane.
4. From the menu options, select **New Custom Model** .



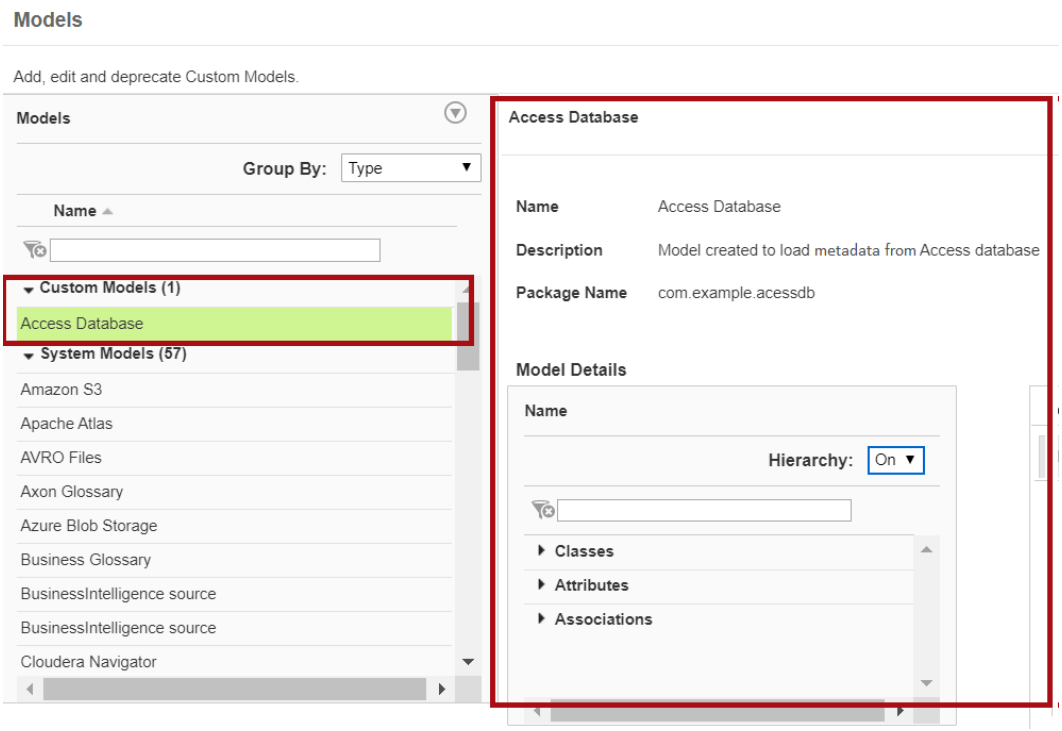
The **New Model** dialog box appears.



5. Click Browse and select the **Access Database XML** file that you had created and saved.



After the model is registered with Catalog Administrator, it appears in the **Models** pane as shown in the following image:

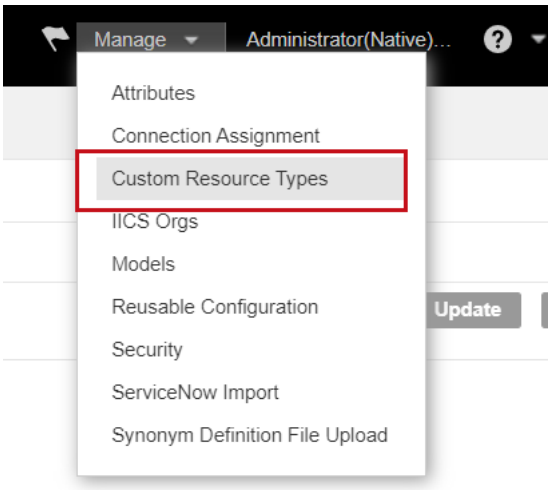


Lesson 2: Create a Custom Resource Type

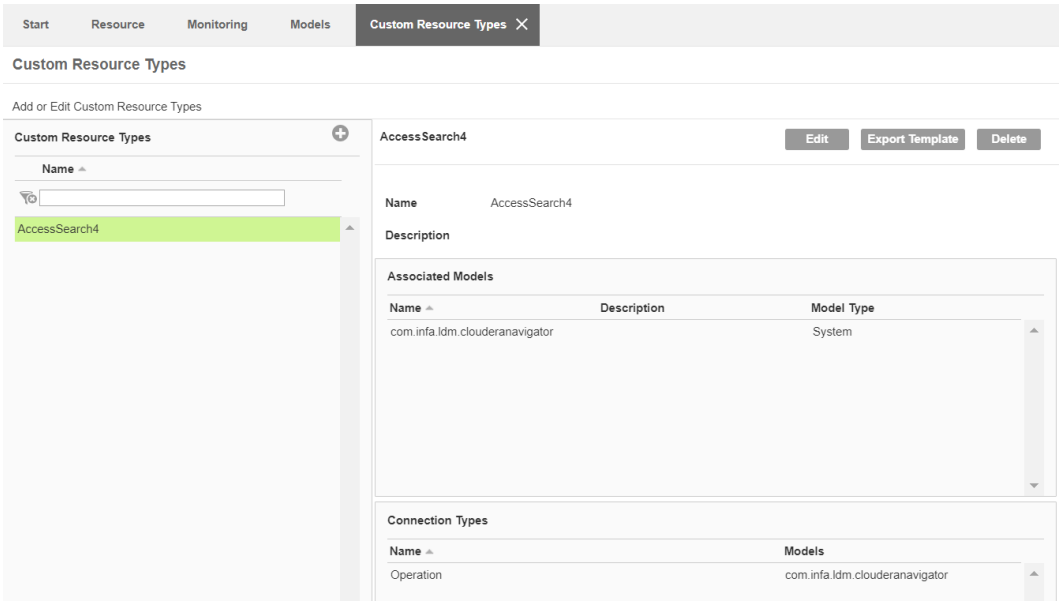
Create a custom resource type based on the custom model that you defined. When you create custom resource type, you define the models that you want to associate with the custom resource type. In addition, you also specify the associations in the selected models that you want to associate with the custom resource type.

In this lesson, we create the **Access Database** resource type and associate it with the **com.example.acessdb** custom model we created in the previous lesson.

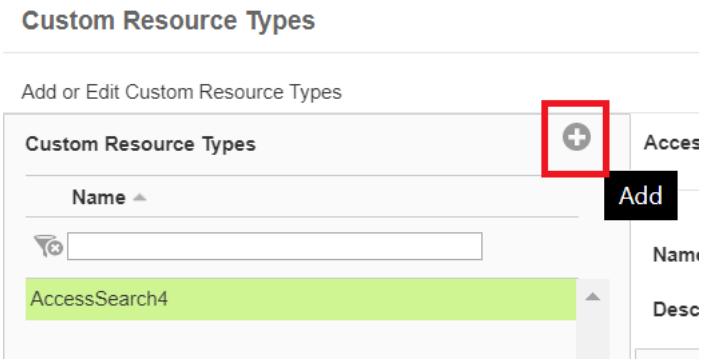
1. In the Catalog Administrator header area, click **Manage** > **Custom Resource Type** as shown in the following image:



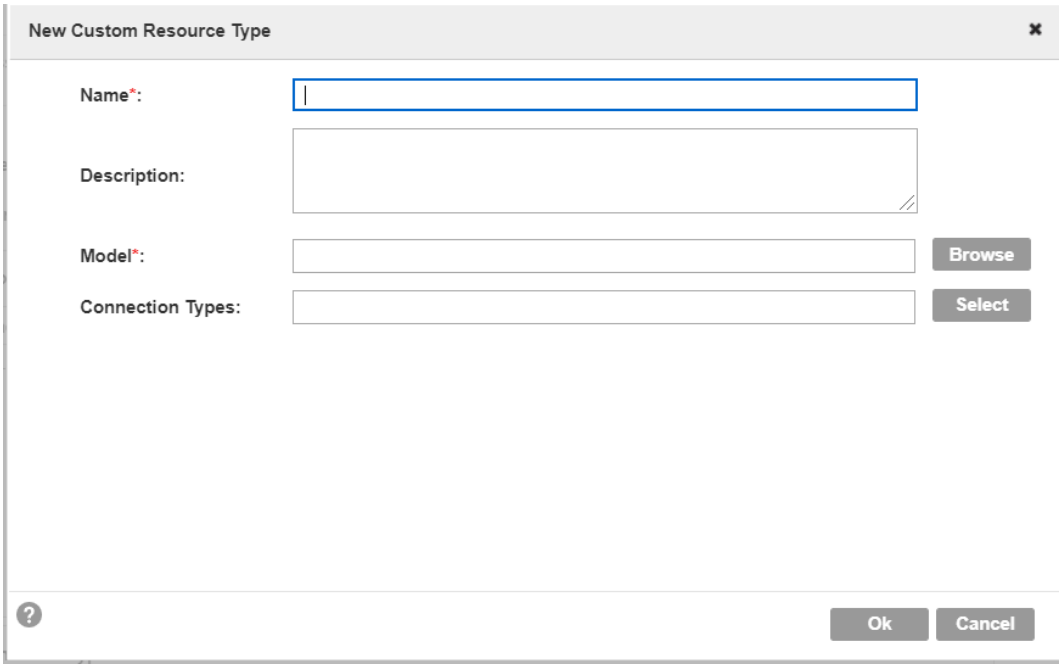
The **Custom Resource Types** page appears as shown in the following image.



2. Click the **Add** button.



The **New Custom Resource Type** dialog box appears.



3. Enter `Access Database` as the name.

   Enter a name that starts with a letter. The name can include letters, numerals, underscore (_), hyphen (-), period (.), and white space.

4. Enter `A custom resource type for Microsoft Access.` as the description.

5. Click **Browse** to find the model that you created in "Lesson 1: Create the Custom Model" on page 3.

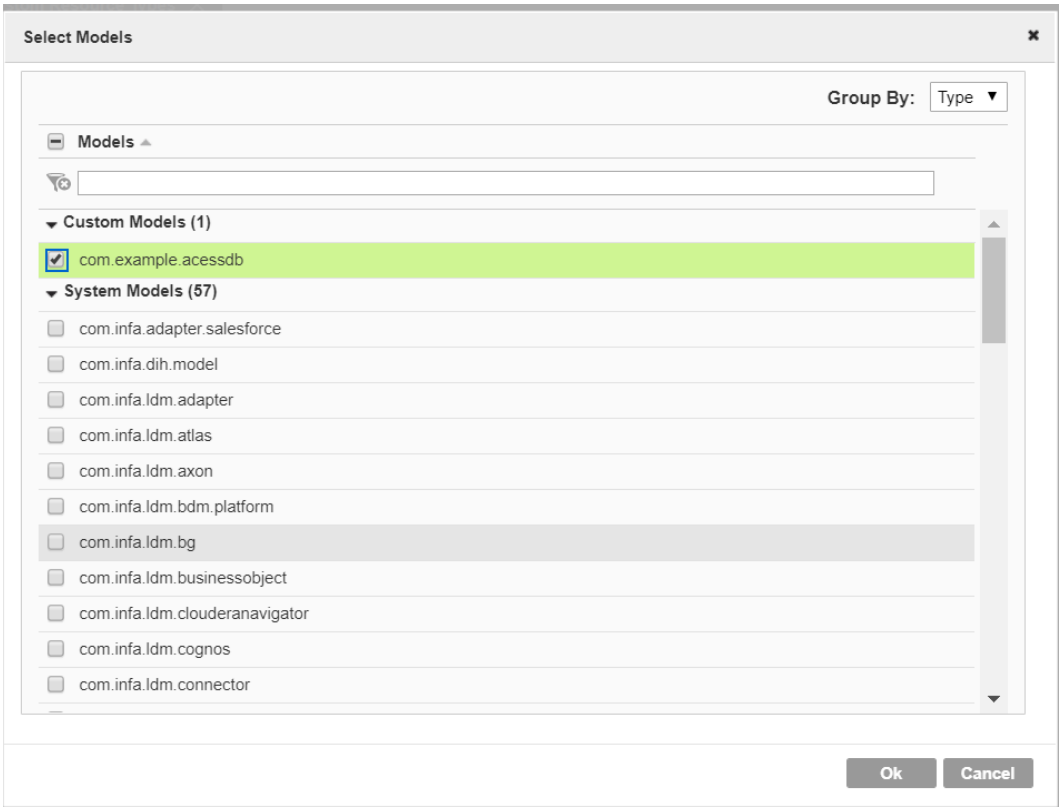
   The **Select Models dialog** box appears.

6. Select the **com.example.acessdb** custom model, and then click **OK** as shown in the following image:

7. Click **Select**.

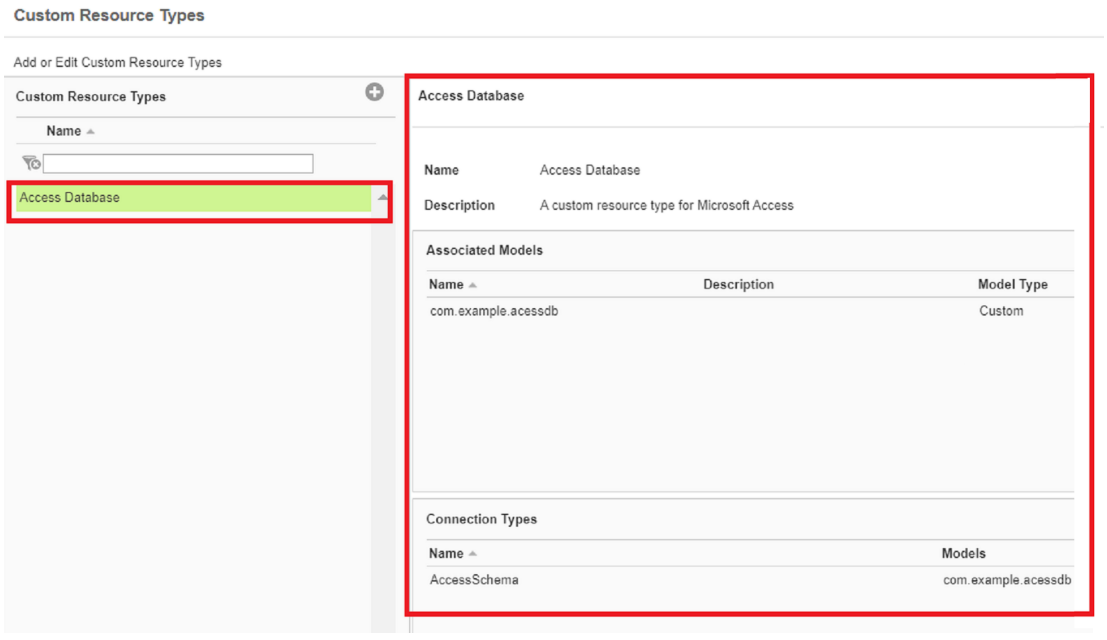
   The **Select Connection Types** dialog box appears.

8. Select the classes that you want to assign as connection endpoints in the custom resource type, and then click **OK** as shown in the following image:



9. Click **OK**.

The **Access Database** custom resource type appears on the **Custom Resource Type** page as shown in the following image:

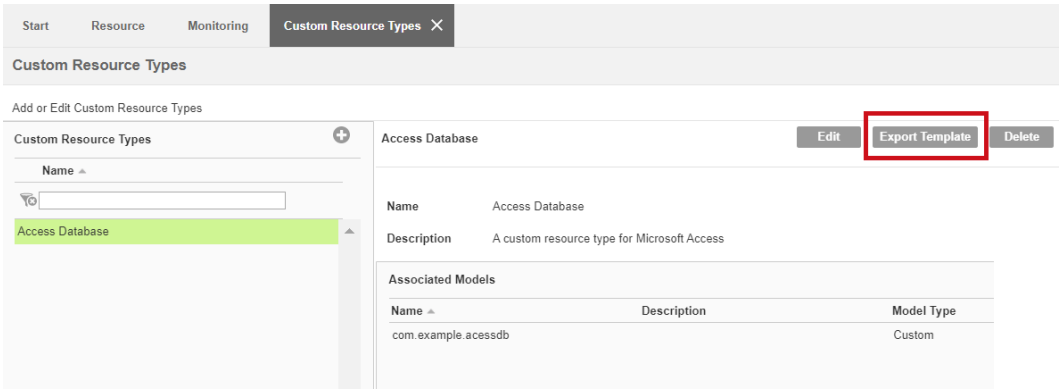


Lesson 3: Export the Custom Resource Type Template

You can use the custom resource type template to add the object details of a specific class defined in the model. Enterprise Data Catalog uses the details entered in the template to ingest metadata into the catalog.

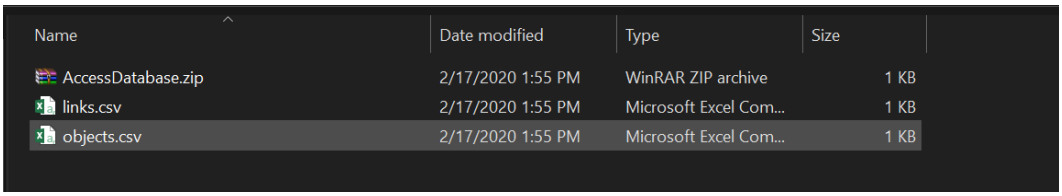
In the lesson, we will learn how to export the **Access Database** export template.

1. From the **Custom Resource Type** page, select **Access Database** custom resource type.
2. Click **Export Template** as shown in the following image:



   The custom resource type template is downloaded to your machine as a .zip file.

3. Extract the files in the .zip file to your machine as shown in the following image:



   The AccessDatabase.zip file contains the following .csv files:

   - **objects.csv.** Use this file to enter data based on the class details from the data source.
   - **links.csv.** Use this file to enter association details of objects from the data source.

Lesson 4: Enter the Association Details

After you extract the .zip file, you must enter the association and class details from the data source in the .csv files.

In this lesson, we will learn how to enter the association details in the links.csv file.

1. Open the **links.csv** file in any text editor.
2. Enter the association details for the data in the custom data source in the following format: `association, fromObject, toObject`

In this lesson, we will enter the association details in the **links.csv** file as shown in the following table:

Association	fromObjectIdentity	toObjectIdentity
core.ResourceParentChild	ACCESS	HERMES
core.ResourceParentChild	ACCESS	WEMBLEY
com.example.acessdb.AccessSchemaAccessTable	HERMES	HERMES/CUSTOMER
com.example.acessdb.AccessTableAccessTableColumn	HERMES/CUSTOMER	HERMES/CUSTOMER/ CUSTOMER_ID
com.example.acessdb.AccessTableAccessTableColumn	HERMES/CUSTOMER	HERMES/CUSTOMER/ CUSTOMER_NAME
com.example.acessdb.AccessSchemaAccessView	HERMES	HERMES/V_CUSTOMER
com.example.acessdb.AccessViewAccessViewColumn	HERMES/V_CUSTOMER	HERMES/V_CUSTOMER/ V_CUSTOMER_ID
com.example.acessdb.AccessViewAccessViewColumn	HERMES/V_CUSTOMER	HERMES/V_CUSTOMER/ V_CUSTOMER_NAME
core.DataSourceDataFlow	HERMES	WEMBLEY
core.DataSetDataFlow	HERMES/CUSTOMER	HERMES/V_CUSTOMER
core.DirectionalDataFlow	HERMES/CUSTOMER/ CUSTOMER_ID	HERMES/V_CUSTOMER/ V_CUSTOMER_ID
core.DirectionalDataFlow	HERMES/CUSTOMER/ CUSTOMER_NAME	HERMES/V_CUSTOMER/ V_CUSTOMER_NAME

3. Save the **links.csv** file with the updates.
4. Replace the **links.csv** in the **AccessDatabase.zip** file with the updated **links.csv** file.

Lesson 5: Enter the Class Details

After you extract the .zip file, you must enter the association and class details from the data source in the .csv files.

In this lesson, we will learn how to enter the class details in the objects.csv file.

1. Open the **objects.csv** file in any text editor.
2. Enter the class details for the data in the custom data source in the following format: `class, identity, core.name, core.description, attributes of the class`

   The class, identity, and core.name are mandatory fields in the **objects.csv** file.

In this lesson, we will enter the class details in the **objects.csv** file as shown in the following table:

Class	identity	core.name	core.description	com.example.accessdb.DateCreated	com.example.accessdb.Datatype
com.example.accessdb.AccessSchema	HERMES	HERMES	This is a schema	7/21/2020	String
com.example.accessdb.AccessSchema	WEMBLEY	WEMBLEY	This is a schema	7/21/2020	String
com.example.accessdb.AccessTable	HERMES/CUSTOMER	CUSTOMER	This is a table	7/21/2020	String
com.example.accessdb.AccessTableColumn	HERMES/CUSTOMER/CUSTOMER_ID	CUSTOMER_ID	This is a table column	7/21/2020	String
com.example.accessdb.AccessTableColumn	HERMES/CUSTOMER/CUSTOMER_NAME	CUSTOMER_NAME	This is a table column	7/21/2020	String
com.example.accessdb.AccessView	HERMES/V_CUSTOMER	V_CUSTOMER	This is a view	7/21/2020	String
com.example.accessdb.AccessViewColumn	HERMES/V_CUSTOMER/V_CUSTOMER_ID	V_CUSTOMER_ID	This is a view column	7/21/2020	String
com.example.accessdb.AccessViewColumn	HERMES/V_CUSTOMER/V_CUSTOMER_NAME	V_CUSTOMER_NAME	This is a view column	7/21/2020	String

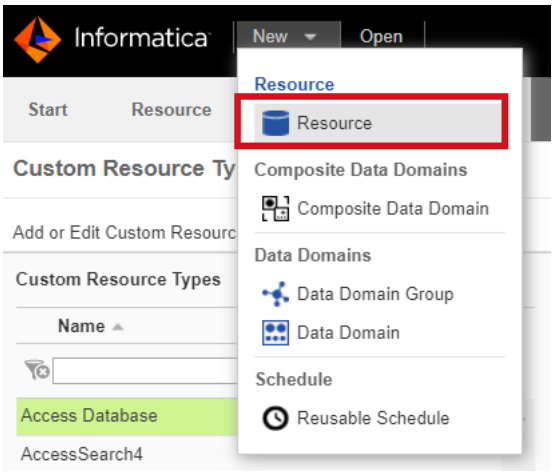
3. Save the **objects.csv** file with the updates.
4. Replace the **objects.csv** file in the **AccessDatabase.zip** file with the updated **objects.csv** file.

Lesson 6: Create the Custom Resource

After you create a custom resource type, create a custom resource based on the custom resource type.

In this lesson, we will learn how to create the **Reporting_AccessDB** custom resource using the **Access Database** custom resource type that we created.

1. Click **New** > **Resource** as shown in the following image:



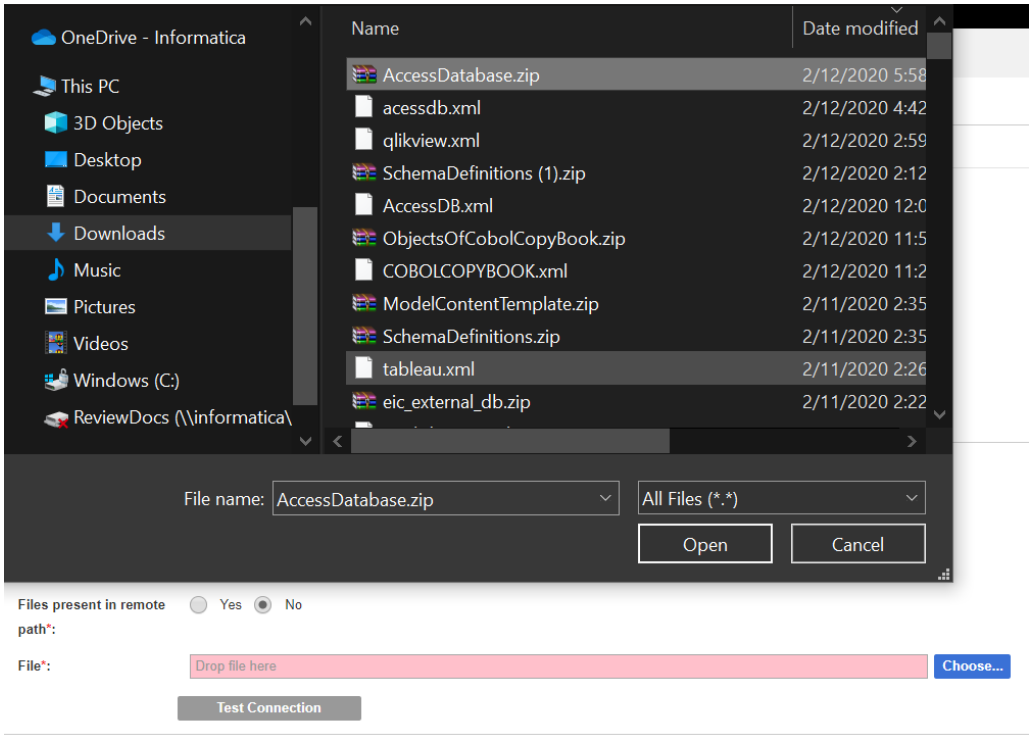
   The **New Resource** wizard appears.

2. On the **General** tab, enter the following details:

Option	Value
Name	Reporting_AccessDB
Description	Database used for reporting on order and sales.
Resource Type	Access Database

3. Enter the following connection properties:
   - ETL Resource. Select **No**. Select **Yes** if you are configuring an ETL resource to view the transformation lineage.
   - Run Prescript. Select **No**. Select **Yes** if you have automated the process to collect metadata from the data sources and generate the .zip file using a script or a sequence of commands.
   - Files present in remote path. Select **No**. Select **Yes** if you want to select the .zip file that includes the custom metadata from a remote file location.

- File. Click **Choose** to select the .zip file that you created for the custom data source as shown in the following image:



Provide the metadata that you want to extract from the source as a .zip file when you configure the custom resource.

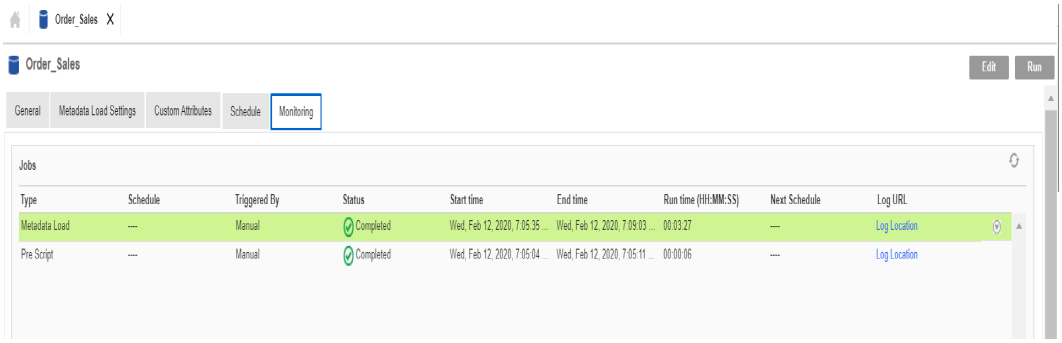
4. Click **Test Connection**.
5. Click **Next** to move to the **Metadata Load Settings** tab.
6. On the **Metadata Load Settings** tab, enter the following details:

Option	Value
Enable Source Metadata	Select this option to extract metadata from the .zip file.
Case Sensitive	Select this option to retain the case sensitivity of the source metadata.
Memory	`Low`
Track Data Source Changes	Select this option to track data source changes.

7. Click **Next** to move to the **Custom Attributes** tab.
   Configure the attribute settings. You can select the custom attributes that you want to associate with the resource.
   In this lesson, we will not assign custom attribute values for the resource.
8. Click **Next** to move to the **Schedules** tab.
   Optionally, select schedules for source metadata load and profiling metadata. You can create a global schedule if required.
   In this lesson, we will not configure schedules for the resource.

9. Click **Save and Run**.

   The custom resource status appears on the **Monitoring** page as shown in the following image:



Lesson 7: View the Custom Metadata Resource and Assets in the Catalog

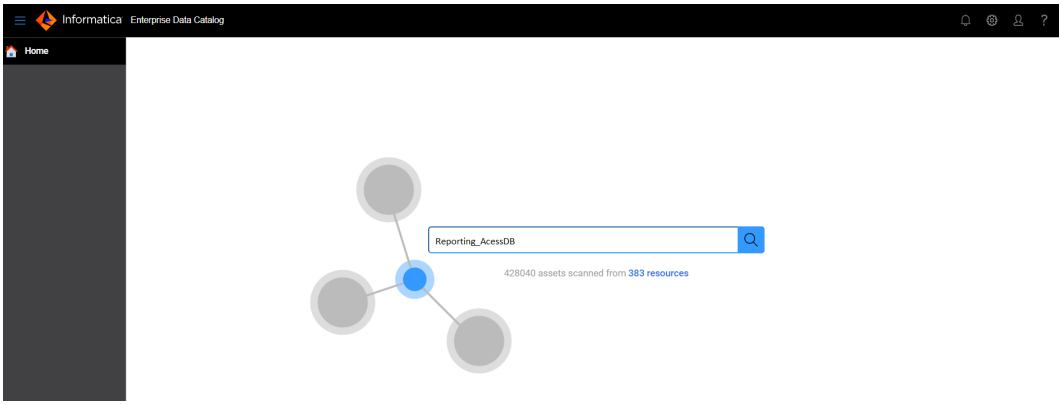
After the **Reporting_AccessDB** custom resource run is complete, you can search for the custom metadata in the catalog.

In this lesson, we will search for the **Reporting_AccessDB** resource and view the asset details.

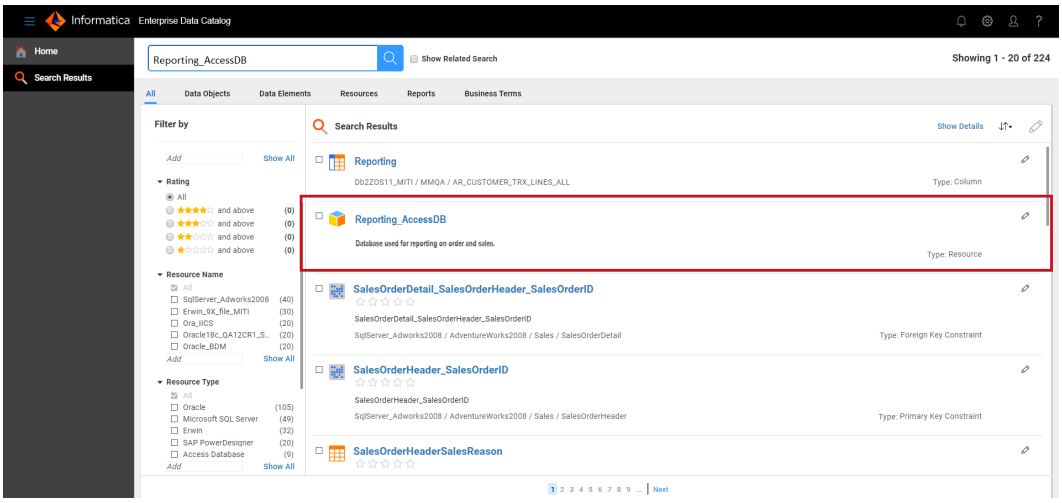
1. Log in to Enterprise Data Catalog.

   The Enterprise Data Catalog search page appears.

2. Type **Reporting_AccessDB** in the search box and click the search icon as shown in the following image:

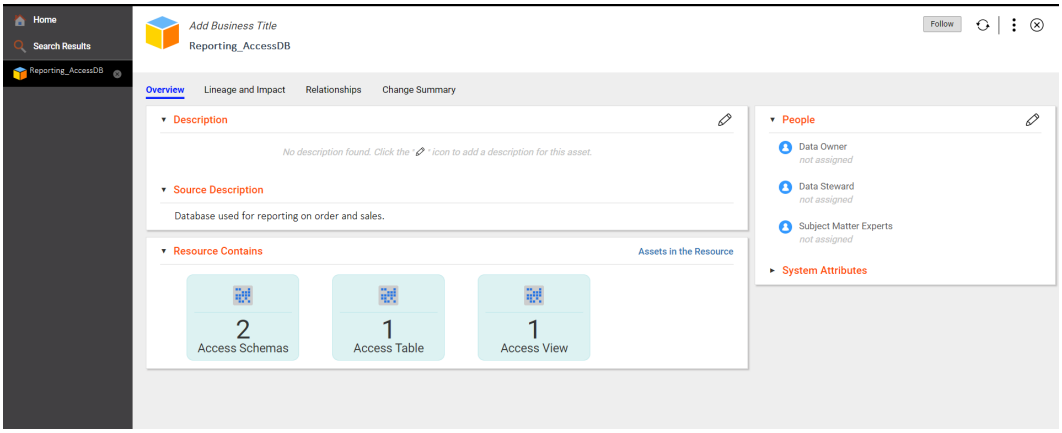


The matching results appear as shown in the following image:



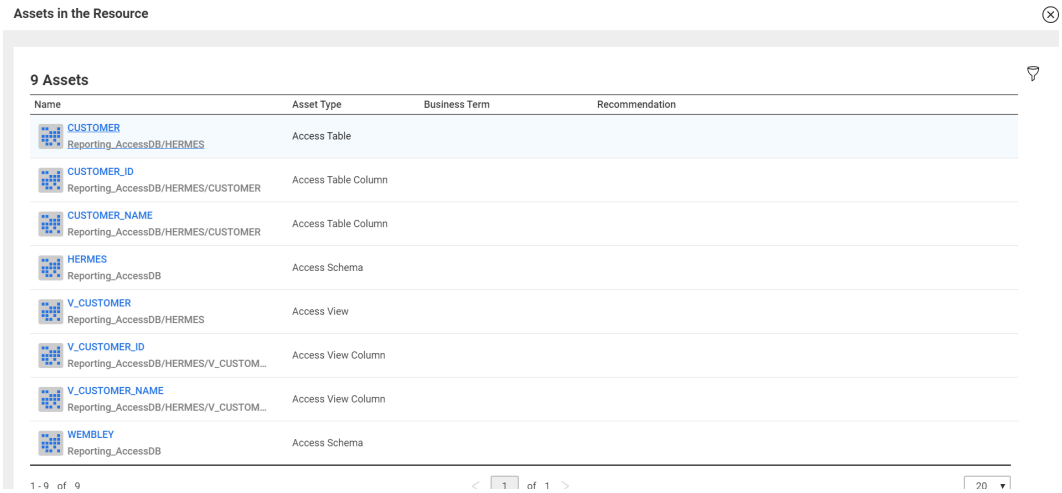
3. Click the resource name link from the search results.

   The **Reporting_AccessDB** resource **Overview** tab appears as shown in the following image:



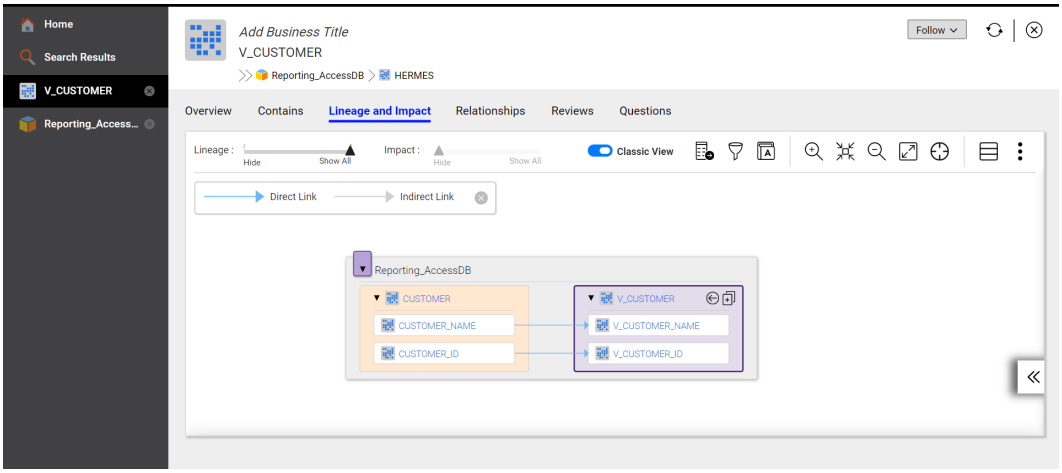
4. On the **Overview** tab, click **Assets in the Resource** link to view the list of assets in the resource

   The **Assets in the Resource** page appears as shown in the following image:



5. Click the **V_CUSTOMER** asset name link and open the **Lineage and Impact** tab.

   The **Lineage and Impact** tab appears as shown in the following image:



You can now view the ingested metadata from the Microsoft Access Database in Enterprise Data Catalog.

Author

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